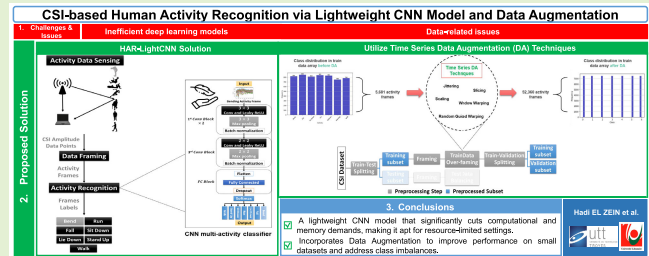


CSI-Based Human Activity Recognition via Lightweight CNN Model and Data Augmentation

Hadi El Zein¹, Farah Mourad-Chehade², and Hassan Amoud³

Abstract—Human activity recognition (HAR) involves detecting users' actions through the analysis of sensor and other data types using AI. While cameras and sensor data are often limited by privacy concerns and technical limitations, researchers have explored device-free solutions. These solutions leverage Wi-Fi signals, which are notably influenced by human movements at the level of channel state information (CSI). This article proposes HAR-LightCNN, a CSI-based HAR solution. The main component of this solution is a deep lightweight convolutional neural network (CNN), notable for its reduced computational demands. The network achieves a balance between depth and trainable parameters, aiming at facilitating real-time activity recognition without compromising performance. We enhance the model's generalization capabilities using time series data augmentation (DA) techniques, which help address the small-sized dataset and class-imbalance problems. Upon evaluation with unseen testing data, our method demonstrates high accuracy in single-user activity recognition, surpassing existing state-of-the-art approaches.

Index Terms—Channel state information (CSI), convolutional neural networks (CNNs), data augmentation (DA), deep learning (DL), human activity recognition (HAR), residential healthcare applications.



I. INTRODUCTION

HUMAN activity recognition (HAR) is the study of identifying and predicting single or multiple users' actions, such as periodic body parts movements, predefined actions, and spontaneous activities. It is gaining prominence in healthcare, robotics, and Internet-of-Things (IoT) applications, notably in residential healthcare. HAR is crucial for monitoring children, the elderly, patients, and the disabled, allowing for timely interventions in emergencies such as falls or abnormal respiration [1], [2].

Manuscript received 11 May 2024; accepted 5 June 2024. Date of publication 19 June 2024; date of current version 1 August 2024. This work was supported in part by the University of Technology of Troyes and in part by Lebanese University. The associate editor coordinating the review of this article and approving it for publication was Prof. Fuqiang Gu. (Corresponding author: Hadi El Zein.)

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Digital Object Identifier 10.1109/JSEN.2024.3414168

There is a great deal of research on HAR using various activity measurement technologies, including cameras [3], [4], sensors [5], [6], [7], and wireless communications (self-powered sensors) [8], [9]. However, using cameras and sensors in a residential environment poses many limitations. For example, cameras require good lighting and can only detect actions within the line of sight, while sensor data can be inaccurate if not worn properly. In light of these challenges, the proposed approach uses the Wi-Fi wireless communication technology as a through-wall detection solution with fewer privacy concerns than cameras. Furthermore, Wi-Fi routers are affordable, power-efficient, insensitive to lighting conditions, easy to implement, and have a long range of connectivity [1], [2], [10].

Wi-Fi signals, vulnerable to absorption, reflection, diffraction, scattering, and interference, change when encountering obstacles such as furniture or moving humans. These interactions create a Wi-Fi channel with multipath overlaid signals. Analyzing how moving objects impact signals helps in recognizing actions. This analysis uses channel state information (CSI), detailing Wi-Fi signal journey from transmitter to receiver, encapsulating reflection, scattering, fading, and power decay effects [10], [11]. CSI data collection happens at the receiver, where AI identifies the raw CSI data. Building these AI systems involves various techniques; although classical

machine learning (ML) methods were previously used, the deep learning (DL) sector of ML now surpasses them in performance [12], [13].

Previous works have explored different DL architectures and data modalities to optimize accuracy and robustness in the “Wi-Fi CSI-based HAR” research domain. For example, one approach involved transforming CSI data into RGB images to feed into a two-dimensional CNN (2D-CNN) model, while also experimenting with an attention-based bidirectional long short-term memory (ABLSTM) model for the same dataset, illustrating the diversity in handling signal CSI data [14]. Others have focused on sequence modeling using raw CSI data with LSTM models for multiactivity classification, highlighting the strengths of sequential data processing in capturing temporal dependencies [15]. Moreover, innovative approaches such as semi-supervised generative adversarial network (GAN) have been introduced to enhance performance on unseen users by generating synthetic training samples, indicating a shift toward adaptable and user-independent models [16]. A hybrid model integrating convolutional neural networks (CNNs) and BLSTMs has also been highlighted, emphasizing the benefits of combining spatial and temporal features for robust HAR [17]. Building on these developments, our previous work [18] explored HAR using Wi-Fi CSI, where we introduced a deep CNN for accurate activity classification and used data augmentation (DA) to address the small-sized dataset limitation.

This article introduces HAR-LightCNN, a methodology based on a lightweight CNN designed for recognizing seven single-user human activities (bending, falling, lying down, running, sitting down, standing up, and walking) using Wi-Fi CSI data, without the need for wearables. Key for monitoring patients, especially the elderly, our approach addresses the prevalent limitation of high model complexity found in DL solutions, making it suitable for deployment in resource-constrained environments. The lightweight design of HAR-LightCNN reduces both computational complexity and memory usage while maintaining robust performance, positioning it as an ideal candidate for real-time activity recognition applications. Moreover, we use time series DA techniques, selected through empirical research, to address the small-sized dataset and class-imbalance limitations, thereby enhancing the model’s performance and generalizability across minority classes. HAR-LightCNN’s performance is confirmed using two testing subsets (prepared from papers’ [14] and [15] datasets) that serve as unified benchmarks for comparing the proposed approach against state-of-the-art ones, demonstrating its superior performance. Compared with the 2D-CNN model proposed by Moshiri et al. [14], our model is approximately $21.32\times$ smaller in terms of parameters, $2.77\times$ less complex in terms of floating-point operations (FLOPs), and 8.7% more precise in terms of accuracy, showcasing both improved efficiency and effectiveness. Compared with our previous work [18], this article presents a thorough literature review for “Wi-Fi CSI-based HAR,” maintaining the deep model architecture while emphasizing its lightweight nature. This article expands our methodology by incorporating DA techniques in addressing the class-imbalance problem, another common

data-related concern. A new model instance, trained on a new dataset alongside the existing one, showcases our approach’s effectiveness. The detailed methodology and results analysis underscore the importance of nuanced approaches to data-related concerns. In addition, introducing new benchmarks from these datasets marks our work as foundational for future Wi-Fi CSI-based HAR research.

This article is structured as follows. Section II presents an overview of the existing Wi-Fi CSI-based solutions for HAR. Section III describes the principle of Wi-Fi CSI-based HAR and the two used datasets in the proposed HAR-LightCNN solution. Section IV digs into the various elements of this solution. The results of the experiments are presented and discussed in Section V. Finally, Section VI concludes and lays out future research directions.

II. RELATED WORKS

Various solutions have been proposed to address the HAR problem, with a growing interest in sensor-less and camera-less approaches. Wi-Fi CSI-based approaches have received significant attention, using traditional ML or DL techniques. In this section, we will review several ML-based approaches in Section II-A and DL-based approaches in Section II-B.

A. HAR ML-Based Approaches

In ML-based approaches, WiFall [19], a fall detection system, uses the local outlier factor (LOF) algorithm to distinguish the activity signals from the anomaly ones and classifies falling from other activities using a one-class support vector machine (SVM). Another system, CARM [20] uses principal component analysis (PCA) to discover the correlations across different CSI streams and discrete wavelet transform (DWT) to extract three types of features for HAR using hidden Markov models (HMMs). Next, RT-Fall [21], a real-time fall detection system, uses 1-D linear interpolation and bandpass filter (BPF) to filter the input CSI data signal and SVM to classify falling from fall-like activities using six extracted features. Finally, WiHACS [22], an HAR system, performs on the collected data a bunch of preprocessing techniques which are 1-D linear interpolation, Hampel filter, 1-D DWT, and detrending. Then, it uses a multiclass SVM for multiactivity classification based on statistical, FFT, and correlation features extracted from the CSI amplitude and phase data.

B. HAR DL-Based Approaches

In DL-based approaches, Moshiri et al. [14] proposed an HAR approach using 2D-CNN and ABLSTM. They collected CSI amplitude data signals for seven daily single-user activities in an indoor environment and used the MATLAB `pcolor()` method to convert them into RGB images. These images were then resized and fed into the CNN model without denoising. In another approach, they trained the BLSTM model on the raw CSI amplitude data. Yousefi et al. [15] carried out the multiactivity classification task using an LSTM model trained on raw CSI amplitude data collected for seven daily single-user activities in an indoor office area. Xiao et al. [16] proposed a semi-supervised GAN model named CsiGAN to improve

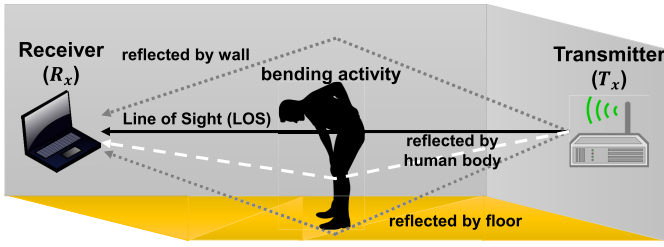


Fig. 1. Wi-Fi signal propagation.

activity recognition on new users. The model uses a complement generator that generates synthetic samples from limited unlabeled data of these users to train a modified discriminator. They trained distinct instances of their model on raw CSI data from two datasets of papers [23] and [24]. Wang et al. [25] proposed an HAR and indoor localization approach using a one-dimensional Residual net (ResNet1D) model trained on raw CSI amplitude data collected at 16 indoor locations for six single-user predefined gestures. Shi et al. [17] developed an HAR scheme named MatNet-eCSI, which consists of collecting CSI data, preprocessing it, and recognizing activities using a hybrid CNN-BiLSTM model. The model is trained using a dataset that includes samples from a previously seen environment, as well as at least one sample from the new testing environment. Finally, Chen et al. [26] carried out the multiactivity classification task using distinct instances of an ABLSTM model trained on raw CSI amplitude data collected for seven single-user spontaneous activities in two indoor environments and the dataset from the paper [15].

III. MATERIALS

A. Wi-Fi CSI-Based HAR

Routers operate on 2.4- or 5-GHz frequency ranges, divided into channels for data transmission. Obstacles affect Wi-Fi, creating multipath overlaid signals. By analyzing how moving objects influence signals, we can identify activities. This is done using fine-grained CSI, which describes Wi-Fi signal propagation, including reflection, scattering, fading, and power loss over distance (see Fig. 1). CSI, measured at the receiver for each channel, captures signal variations in amplitude and phase over time or space [10], [11].

Orthogonal frequency-division multiplexing (OFDM) enables simultaneous sharing of a communication channel among wireless network nodes. Used in standards like Wi-Fi 802.11ac, 4G, and 5G, OFDM breaks channels into overlapping subcarriers for increased data rates and efficient bandwidth use. Subcarrier fading affects channel dynamics, and CSI can catch the changes occurring at a single frequency. Collecting CSI data involves hardware and software, such as the Intel 5300 Wi-Fi network interface card (NIC) with Linux 802.11n CSI Tool [27], Atheros Wi-Fi NICs with Atheros-CSI-Tool [28], and Raspberry Pi with Nexmon CSI Extractor [29]. These tools enable the extraction of detailed physical layer wireless communication information. Accordingly, CSI is measured for each subcarrier of a particular channel, and it is represented by a complex number

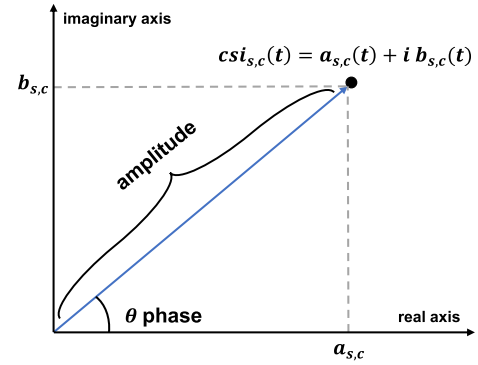


Fig. 2. CSI at time t for subcarrier s of channel c .

given by the equation

$$csi_{s,c}(t) = a_{s,c}(t) + i b_{s,c}(t) \quad (1)$$

where $s \in \mathcal{S}$ is the subcarrier index, $c \in \mathcal{C}$ is the channel index, $a_{s,c}(t)$ is the real part of the CSI measured at time t for subcarrier s of channel c , and $b_{s,c}(t)$ its imaginary part. Fig. 2 represents the amplitude and phase of a measured $csi_{s,c}(t)$. Activities in Wi-Fi field have a bigger impact on received signal amplitude compared with phase. This work uses CSI amplitude data points, given by $x_{s,c}(t) = (a_{s,c}(t)^2 + b_{s,c}(t)^2)^{1/2}$, as inputs for HAR [11], [28], [30].

B. First CSI Dataset

This article uses the publicly available activity CSI dataset collected by Moshiri et al. [14] to illustrate the approach. The data were collected using a transmitter and receiver having a Nexmon CSI extractor [29], with a single channel and 52 subcarriers. In an indoor experimental environment, the receiver collected the CSI amplitude data for seven daily activities (bending, falling, lying down, running, sitting down, standing up, and walking), at a sampling rate of 200 samples/s. Three participants of varying ages performed each activity 20 times, and the collected time series data for each activity trial were exported into 420 CSV files (60 files per activity). Each file contains 52 columns for the 52 subcarriers and 600–1100 rows, depending on the activity duration and sampling rate.

C. Second CSI Dataset

Another publicly available activity CSI dataset collected by Yousefi et al. [15] is also used. The data were gathered using a transmitter and receiver with an Intel 5300 Wi-Fi NIC and the Linux 802.11n CSI Tool [27]. The NIC was equipped with three antennas, while the transmitter had only one, resulting in three channels with 30 subcarriers each. The dataset also includes the seven daily activities, listed above, recorded at a sampling rate of 1000 samples/s. Six participants performed each activity a varying number of times, leading to a total of 557 signal data files (around 80 files per activity). Each activity trial CSV file contains 90 columns, representing the 30 subcarriers across three channels, and around 20 000 rows, corresponding to the duration of the trial.

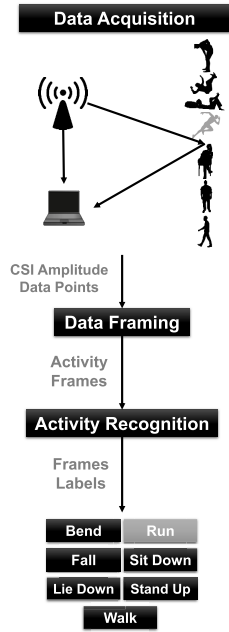


Fig. 3. HAR-LightCNN three-phase design.

IV. METHODS

A. HAR-LightCNN Solution

The proposed solution, termed HAR-LightCNN is composed of three main phases: data acquisition, data framing, and activity recognition. The architecture is illustrated in Fig. 3. During data acquisition, the transmitter sends Wi-Fi signals and the receiver collects in real-time the CSI amplitude data $x_{s,c}(t)$, $s \in S$, $c \in C$. In the data framing phase, the collected data are grouped into frames of a fixed duration of 1 s. These frames, known as activity time series data frames, are then processed to comply with the input requirements of the next phase. This may include downsampling or oversampling. In the final phase of activity recognition, our designed and trained classifier, detailed in Section IV-D, categorizes the activity frames into one of the seven considered daily activities: bending, falling, lying down, running, sitting down, standing up, and walking. The subsequent sections will detail the preprocessing procedures applied to the datasets used for training, as well as the architecture of the classifier.

B. Preprocessing of Datasets

The first dataset from [14] undergoes preprocessing for our CNN multiactivity classifier training and testing. We split the data into 85% training and 15% testing, yielding 357 and 63 files, respectively. The data are processed using overlapped framing with a frame size of 200 (matching the sampling rate) and a slide size of 40, resulting in 1-s activity snapshots. This creates 5681 training frames and 994 testing frames of size (200, 52). To mitigate class imbalance in the testing subset, frames are discarded to match the minority class (sitting down) at 128 frames, ensuring unbiased evaluation. These steps reflect real-world model performance on unseen data. The training subset, without further division due to its small size, undergoes time series DA (detailed in Section IV-C) and

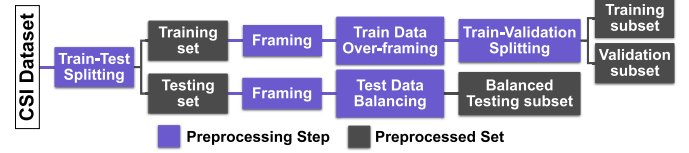


Fig. 4. Dataset preprocessing steps.

then a train-validation split. Two experiments are conducted, one with the conventional holdout method (train-validation split of 85%–15%) and another using fivefold k -fold cross-validation (CV) [13].

The second dataset from [15] is imbalanced, and the same preprocessing steps outlined for the first dataset are applied to it. It is worth noting that like the first dataset, this one also uses 1-s snapshots, but with a downsampling factor of 2 to prevent memory errors and decrease computational expenses. Fig. 4 provides an overview of the preprocessing steps applied to both the datasets.

C. Data Augmentation

The primary challenges with real datasets are their small size and class imbalance, which hinder the effectiveness of DL techniques. Small datasets can cause overfitting, leading to poor model performance. Moreover, many ML techniques assume a balanced number of training frames for each class, and imbalanced datasets can negatively affect classification, especially for underrepresented classes. To overcome these issues, we suggest using two types of DA techniques: random-transformation-based and pattern-mixing-based [12], [31], [32].

Inspired by image DA, the random-transformation-based DA techniques generate new frames $\mathbf{x}'$ by applying transformations to the existing ones $\mathbf{x} = (\mathbf{x}_1, \dots, \mathbf{x}_T)$. We apply various types of transformations to time series data frames, such as those related to magnitude, time, or frequency domains. These include jittering and scaling for generating extra patterns in the magnitude domain, and slicing and time warping in the time domain. As for jittering, it adds noise, usually Gaussian noise, to the time series frames, as described by the equation

$$\mathbf{x}' = (\mathbf{x}_1 + \boldsymbol{\epsilon}_1, \dots, \mathbf{x}_t + \boldsymbol{\epsilon}_t, \dots, \mathbf{x}_T + \boldsymbol{\epsilon}_T) \quad (2)$$

where $\boldsymbol{\epsilon}_t$ is the noise added to each sample (data point) $\mathbf{x}_t$ at time step t , such that $\boldsymbol{\epsilon} \sim N(\mathbf{0}, \sigma_{\epsilon}^2)$. The scaling technique is applied to change the global magnitude of a time series frame by multiplying it with a random scalar value called scaling parameter α , as described by the equation

$$\mathbf{x}' = (\alpha \mathbf{x}_1, \dots, \alpha \mathbf{x}_t, \dots, \alpha \mathbf{x}_T) \quad (3)$$

the scaling parameter α is a random variable with a Gaussian distribution, such that $\alpha \sim N(\mathbf{1}, \sigma_{\alpha}^2)$. As for slicing, it is used to transform time series frames by randomly cropping a window of size W from the samples' time steps, per equation

$$\mathbf{x}' = (\mathbf{x}_{\phi}, \dots, \mathbf{x}_t, \dots, \mathbf{x}_{W+\phi}) \quad (4)$$

where φ is a random integer value $\varphi \in \mathbb{N}$, such that $1 \leq \varphi \leq T - W$. Regarding time warping, it disturbs the time series frames with respect to the time axis. Window warping, random warping, vowel stretching, time stretching, and time perturbation are all time warping methods. We use window warping, a specific method that stretches or contracts a random window of the frame using a multiplier value, often 2 for stretching and 1/2 for contracting [31].

The pattern-mixing-based DA techniques generate new patterns (frames) by combining two or more similar time series frames. They are performed with respect to magnitude, time, or frequency domains. In time-domain mixing, time series frames are combined based on the time axis. Guided warping (GW) is a technique that aligns a reference frame to a teacher frame using dynamic time warping (DTW) to measure similarity. We use random GW (RGW), which is a variant that selects a random intraclass teacher frame [31].

The first dataset of [14] has the small-sized dataset problem. To solve this, four random-transformation-based DA techniques are applied to generate extra new frames on the training frames of each class (activity). These techniques are jittering, scaling, slicing, and window warping. In additionally, RGW, a pattern-mixing-based DA technique is applied to the training frames of the seven classes to generate extra synthetic frames. As a result, the total number of training subset frames increased from 5681 to 52360. These DA techniques were applied to expand the decision boundary of the model and alleviate the overfitting problem, leading to an enhanced generalization ability on unseen data, and ultimately an efficient classifier.

The second dataset of [15] has the class-imbalance problem. To address this, we use time series DA techniques to augment low-instance classes. We first identified the majority class having the most number of frames and then determined the number of additional frames needed for each class to reach that level. On the training frames of each class, we applied multiple random-transformation-based techniques to generate new frames. In addition, we used RGW, a pattern-mixing-based technique, on the training frames of the seven classes to generate additional synthetic frames. As a result, the total number of training subset frames for the second dataset increased from 8497 to 34884, resulting in evenly distributed frames.

Fig. 5 illustrates the application of four random-transformation-based DA techniques on two frames, each derived from real CSI data signals representing bending and lying down activities from the first dataset. Each subplot shows the original frame followed by its augmented versions. These visualizations highlight how each technique modifies the original frame, introducing variability that helps our model generalize better to unseen data. Notably, the transformed frames retain key characteristics of the original activity, ensuring that the augmented data remain relevant for accurate activity classification.

Fig. 6 presents a comparative analysis of original and augmented frames of bending and lying down classes from the first dataset by applying the RGW pattern-mixing-based DA technique. Each subplot showcases the original frame

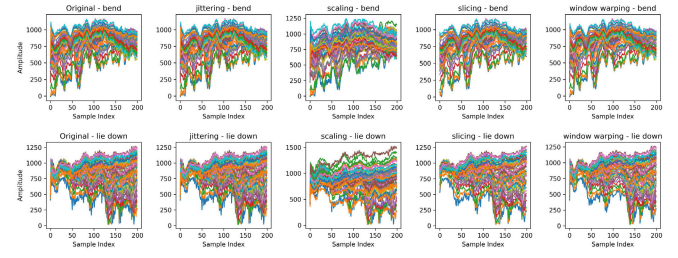


Fig. 5. Augmentation impact of random-transformation techniques.

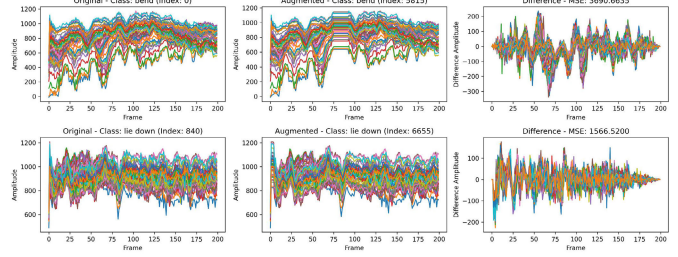


Fig. 6. Augmentation impact of RGW pattern-mixing technique.

alongside its augmented counterpart and their absolute differences. The mean squared error (MSE) values associated with these frames (3690.6635 and 1566.5200) highlight the effectiveness of the RGW technique in introducing subtle yet significant modifications. These MSE values, substantially lower than the square of the standard deviation of the dataset (65987.33), confirm that the alterations to the CSI signal frames are minor and do not drastically distort the underlying data patterns. This level of augmentation ensures the training data, remains representative of real-world variations, fostering the development of robust DL models. The subtle changes introduced are sufficient to add the necessary diversity for the models to classify the exact signals encountered during training and their slight variations, thereby enhancing the models' ability to generalize across real-world scenarios.

D. Proposed CNN Activity Classifier

In Wi-Fi CSI-based HAR, the use of CNNs enhances the performance of the multiactivity classification task. This improvement is due to CNNs' proficiency in extracting features from raw data, managing high-dimensional data, and learning spatiotemporal features in multivariate time series, like the CSI data [11], [12], [33].

The proposed CNN classifier processes the CSI activity frame to identify one of seven activities. It comprises three convolutional (Conv) blocks, a fully connected (FC) block, and an output block. Each Conv block includes Conv2D, LeakyReLU, MaxPooling2D, and BatchNormalization layers. LeakyReLU, with a slope of 0.3, is the activation function. The first two Conv blocks contain 32 filters of 3×3 in the Conv2D layer and a 3×3 pooling window in the MaxPooling2D layer, with "same" zero padding and 2 pixel horizontal and vertical pooling movement. These blocks downsample the input. The BatchNormalization layer normalizes activations for faster model convergence and training. The third Conv block uses 2×2 filters and pooling windows. The FC block has

TABLE I
SUMMARY OF MODEL HYPERPARAMETERS

Hyper-parameter	Chosen Value
Optimizer	Adam
Learning Rate	10^{-4}
Loss Function	Sparse Categorical Cross-Entropy
Metrics	accuracy
Batch Size	32
Number of Epochs	150
EarlyStopping Patience	10 epochs
ReduceLROnPlateau Factor	0.5
ReduceLROnPlateau Patience	5 epochs
ReduceLROnPlateau Min LR	10^{-5}

flatten, dense, and dropout layers; Flatten turns the third Conv block's output into a 1-D array, followed by a dense layer with 64 neurons and ReLU activation. A 30% dropout layer reduces overfitting. The output block has a dense layer with seven neurons and softmax activation, creating a probability distribution for each activity, with the highest score indicating the predicted activity [12], [13]. Fig. 7 shows the overall architecture of our multiactivity classifier.

Our model's initial configurations included using the Adam optimizer with a learning rate of 10^{-4} , and the loss function set to sparse categorical cross-entropy. We trained the model over 150 epochs with a batch size of 32. Various training callbacks were used, including EarlyStopping, which terminates training if no improvement in validation loss is observed for ten epochs. The ReduceLROnPlateau callback adjusts the learning rate by a factor of 0.5 if no improvement in validation loss is seen for five epochs, facilitating finer model adjustments and potentially better performance. The ModelCheckpoint callback was used to save the best-performing model version based on validation accuracy. These hyperparameters and training strategies are summarized in Table I.

This model is considered "lightweight" due to its efficient architecture that substantially reduces the number of parameters compared with known lightweight CNN architectures, making it computationally less expensive. The proposed CNN architecture balances depth (number of layers) and width (number of units or filters per layer) to operate within the constraints of devices with limited computational resources, ensuring real-time processing capabilities. To substantiate our claim of lightweight design, we compared the efficiency of our model against established benchmarks in the field: ShuffleNet [34], SqueezeNet [35] (versions 1.0 and 1.1), and MobileNet (versions V2 [36] and V3 [37]). The comparison focused on efficiency aspect evaluation metrics, which include prediction complexity, quantifying the maximum operations needed to process a single instance by the model as a function of input size, calculated in million FLOPs (MFLOPs) using the keras_flops library designed to assess the computational demands of Keras-based models; the number of parameters, indicating the total quantity of model parameters; and memory requirements, representing the amount of storage needed to hold the model's parameters, quantified in megabytes (MB). Each model (with default configurations) was adapted to process inputs of the same dimensions (200, 52, 1) (tailored to first dataset) to ensure a fair comparison. Table II shows

TABLE II
EFFICIENCY BENCHMARKING OF THE PROPOSED CNN AGAINST
ESTABLISHED LIGHTWEIGHT CNN ARCHITECTURES

Model	Complex. (MFLOPs)	N° Params	Mem. (MB)
ShuffleNet [†]	74.2	1,035,559	3.9
SqueezeNet v1.0 [‡]	224.8	729,607	2.7
SqueezeNet v1.1 [‡]	80.8	724,935	2.7
MobileNetV2 [§]	149.6	2,266,375	8.6
MobileNetV3 [§]	28.6	942,871	3.6
Proposed CNN	53.2	309,511	1.1

[†]Implemented by Haikoitoh (GitHub)

[‡]Implemented by cmasch (GitHub)

[§]Implemented by TensorFlow's Keras API

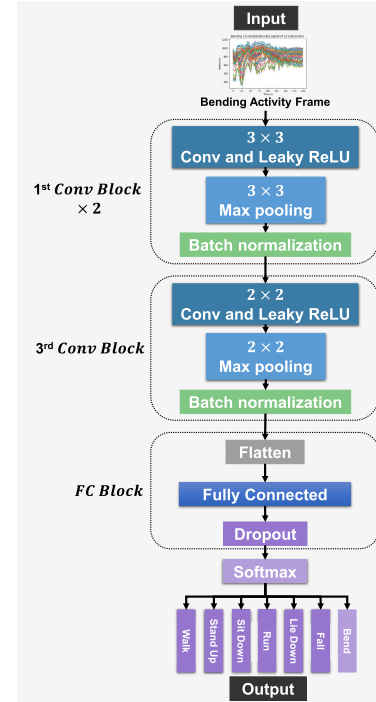


Fig. 7. Architecture of the lightweight CNN multiactivity classifier.

comparison of the efficiency of the proposed CNN model against lightweight CNN models cited in the literature.

The results demonstrate the lightweight nature of the proposed model, showcasing its potential for deployment in environments with stringent memory and computational constraints. By substantially reducing computational complexity and memory usage, our model stands as a viable solution for efficient real-time activity recognition.

V. EXPERIMENTAL RESULTS

Our proposed approach is evaluated using the real datasets from Sections III-B and III-C. The classifier is tested with and without DA and compared against literature approaches. Four evaluation metrics are used: accuracy, $F1$ -Score, training time, and response time. The accuracy measures the percentage of correctly classified instances, the $F1$ -Score is a measure of precision and recall, training time records the model's average training duration per epoch in seconds (s), and response time quantifies the model's prediction latency for each instance within the testing subset in milliseconds (ms).

TABLE III
TESTING RESULTS ON [14] WITHOUT DA

Metric \ Experiment	1 st Experiment				2 nd Experiment					Average
	S_0	S_1	S_2	S_3	F_1	F_2	F_3	F_4	F_5	
Accuracy (%)	89.6	91.8	88.7	88.9	90.4	88.0	89.1	89.7	89.3	89.5
F1-score (%)	89.4	91.7	88.6	88.8	90.2	87.8	88.9	89.6	89.3	89.3
Training Time (s)	1.090	1.166	1.038	1.074	1.361	1.080	1.090	1.109	1.107	1.124
Response Time (ms)	0.263	0.246	0.252	0.245	0.261	0.258	0.257	0.257	0.255	0.255

TABLE IV
TESTING RESULTS ON [15] WITHOUT DA

Metric \ Experiment	1 st Experiment				2 nd Experiment					Average
	S_0	S_1	S_2	S_3	F_1	F_2	F_3	F_4	F_5	
Accuracy (%)	83.6	81.0	83.9	75.6	77.7	81.5	81.1	83.2	82.3	81.1
F1-score (%)	83.5	81.2	83.8	76.1	77.8	81.6	81.3	83.1	82.1	81.1
Training Time (s)	8.523	8.342	8.461	8.352	5.086	5.040	5.037	5.030	5.041	6.546
Response Time (ms)	0.774	0.754	0.756	0.758	0.749	0.751	0.743	0.749	0.745	0.753

TABLE V
TESTING RESULTS ON [14] WITH DA

Metric \ Experiment	1 st Experiment				2 nd Experiment					Average
	S_0	S_1	S_2	S_3	F_1	F_2	F_3	F_4	F_5	
Accuracy (%)	92.7	91.8	93.6	92.8	92.1	93.7	93.7	92.5	93.7	92.9
F1-score (%)	92.7	91.8	93.6	92.8	92.1	93.7	93.7	92.5	93.7	92.9
Training Time (s)	9.311	10.666	10.052	10.105	9.684	10.000	9.954	10.041	10.000	9.979
Response Time (ms)	0.255	0.254	0.252	0.254	0.250	0.246	0.243	0.244	0.245	0.249

A. Evaluation of HAR-LightCNN Without DA

We conducted two experiments on the first dataset [14], using different splitting procedures for the training subset. Table III shows the testing results on the preprocessed unseen testing subset, using holdout splitting repeated four times for the first experiment and CV splitting for the second. The testing accuracies and $F1$ -Scores indicate stable and good performance, with little impact on the splitting method. In addition, training and response times were stable. Our CNN models remained efficient despite limited training data. The same experiments were conducted on the second dataset [15], with the results shown in Table IV. Testing accuracies and $F1$ -Scores indicate stable and good performance, with little impact of the splitting method. Training and response times were also stable, and our CNN models remained efficient despite limited training data.

B. Evaluation of HAR-LightCNN With DA

We repeated the experiments with DA to investigate the impact of increasing the training subset size. Table V shows the testing results on the unseen data from the first dataset [14], using the same two training subsets' splitting methods. The testing accuracies and $F1$ -Scores demonstrate improved classification performance, with an average improvement of 3.4% and 3.6%, respectively. Applying DA to increase the training subset size improved the classifiers' classification performance and generalization ability. Our proposed approach was efficient in enhancing the models' performance. Then, we evaluated the most efficient multiactivity classifier (third fold classifier with DA) focusing on the seven activities, prioritizing the detection of falling activity due to its critical importance. We used a confusion matrix for evaluation, as shown in Fig. 8, and

TABLE VI
CLASSIFICATION REPORT OF MOST EFFICIENT CLASSIFIER ON [14]

Metric \ Class	Precision (%)	Recall (%)	F1-score (%)
Bending	100	98	99
Falling	100	97	98
Lying Down	84	86	85
Running	96	98	97
Sitting Down	87	96	91
Standing Up	93	86	89
Walking	98	96	97

computed precision, recall, and $F1$ -Score for each activity class. Table VI, the classification report, shows comparison of the values of these metrics. For the falling activity, the classifier achieved ideal precision of 100% in identifying only the correct instances for falling, and 97% recall, indicating a strong ability to identify all its correct instances. Also for the other activities, the precision and recall values are great. The $F1$ -Score, a balance of precision and recall, is our chosen metric for assessing overall classification performance, showing excellent results in managing these two aspects of our activities.

We repeated the experiments on the second dataset [15] using DA to examine the effect of balancing the training subset and increasing its size on the models' performance. Table VII illustrates the testing results on unseen data from the second dataset [15], using the same two training subsets' splitting methods. The testing accuracies and $F1$ -Scores indicate improved and stable classification performance, with an average improvement of 3.91% and 3.93%, respectively. The application of DA techniques improved the classification performance and generalization ability of the classifiers. Our

TABLE VII
TESTING RESULTS ON [15] WITH DA

Experiment Metric	1 st Experiment				2 nd Experiment					Average
	S_0	S_1	S_2	S_3	F_1	F_2	F_3	F_4	F_5	
Accuracy (%)	87.3	86.4	84.9	86.5	83.0	84.6	83.3	83.3	85.8	85.01
F1-score (%)	87.1	86.4	85.0	86.5	83.0	84.6	83.3	83.5	85.9	85.03
Training Time (s)	21.285	20.245	34.606	34.352	24.374	24.154	24.156	24.7737	24.7767	25.858
Response Time (ms)	0.801	0.773	0.746	0.757	0.746	0.748	0.742	0.749	0.746	0.756

TABLE VIII
CLASSIFICATION REPORT OF MOST EFFICIENT CLASSIFIER ON [15]

Class \ Metric	Precision (%)	Recall (%)	F1-score (%)
Bed	82	87	84
Falling	94	91	92
Picking Up	94	75	83
Running	88	100	94
Sitting Down	83	76	80
Standing Up	80	82	81
Walking	92	100	96

TABLE IX
F1-SCORE VALUES OF BEST CLASSIFIER ON [15] WITHOUT DA

Class	F1-score (%)	Class Type
Sitting Down	72	Minority
Standing Up	79	Minority
Walking	85	Majority

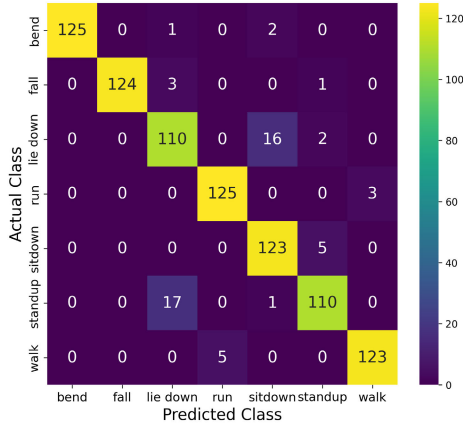


Fig. 8. Confusion matrix for most efficient classifier (with DA) on [14].

proposed approach was effective in enhancing the models' performance. Then, we evaluated the performance of the most efficient multiactivity classifier (with DA). Through the classifier's confusion matrix, displayed in Fig. 9, we computed its precision, recall, and F1-Score for all activities, detailed in Table VIII. For the falling activity, the classifier achieved great precision of 94% and a pretty good recall of 91%. Also for most of the other activities, the precision and recall values are good, and the F1-Scores show a good balance between precision and recall. For comparison, Table IX presents the F1-Scores of the previous best classifier (without DA) for the minority classes (sitting down and standing up) and the walking majority class. Compared with this previous classifier, the F1-Scores for minority classes improved by 8% and 2%, respectively, highlighting enhanced classification of the

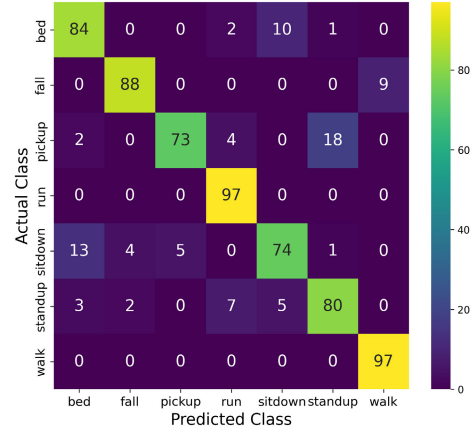


Fig. 9. Confusion matrix for most efficient classifier (with DA) on [15].

classifier on the minority classes' unseen data. Likewise, the F1-Score for the majority class increased by 11%, indicating an overall improvement in the classifier's generalization ability.

In the conducted experiments, DA significantly enhanced the classifier's performance by increasing the diversity and volume of the training subset. However, this augmentation naturally led to an increase in the average training time. Notably, despite the longer training periods, the response times during the testing phase remained consistently unaffected. This stability is crucial for real-time applications such as activity recognition, where prompt decision-making is essential. Therefore, while the augmented training phase requires additional time, an acceptable tradeoff in scenarios where training occurs offline, the consistent response time ensures that the operational efficiency during real-time application is maintained, thereby validating the effectiveness of DA in improving classification accuracy without compromising the model's responsiveness in practical settings.

C. Comparison of HAR-LightCNN With State-of-the-Art

In evaluation, reliability is impacted by certain practices. Preprocessing the CSI dataset by framing it before splitting it into training and testing subsets can cause shared data samples between them. Besides, using the same activity frames for both validation and testing skews evaluation accuracy. These practices lead to results that do not truly represent model performance. To counter this, we prepared two testing subsets as unified benchmarks for reliable and unbiased evaluation.

Our study compared different DL models, as implemented according to the network configurations and hyperparameter settings detailed in [14], against our proposed CNN model

TABLE X
RESULTS ON [14] FOR HAR-LIGHTCNN VERSUS STATE-OF-THE-ART

Metric \ Approach	1D-CNN [14]	2D-CNN [14]	LSTM [14]	BLSTM [14]	HAR-LIGHTCNN
Accuracy (%)	51.6	85.0	74.5	81.7	93.7
F1-score (%)	50.6	84.8	74.3	80.5	93.7
Training Time (s)	0.540	1.146	1.046	1.053	9.954
Response Time (ms)	0.337	0.368	0.468	1.04	0.243
N° Parameters	630,631	6,598,023	106,279	568,407	309,511

TABLE XI
RESULTS ON [15] FOR HAR-LIGHTCNN VERSUS STATE-OF-THE-ART

Metric \ Approach	LSTM [15]	ABLSTM [26]	HAR-LIGHTCNN
Accuracy (%)	68.1	75.6	87.3
F1-score (%)	67.9	75.7	87.1
Training Time (s)	3.140	3.126	21.285
Response Time (ms)	0.916	1.294	0.801
N° Parameters	234,207	629,207	1,284,359

on unseen testing data from their dataset (see Table X). Among these models was the 2D-CNN, which served as a benchmark model and was directly compared to the proposed CNN. Our CNN, streamlined to 309 511 parameters, is approximately 21.32 times smaller than the 2D-CNN, which has 6 598 023 parameters. In addition, it reduces prediction complexity by 2.77 times, requiring about 53.25 MFLOPs compared with 147.75 MFLOPs for the 2D-CNN. It performed significantly better in testing accuracy and *F1*-Score, achieving an 8.7% higher accuracy and *F1*-Score at 93.7%, compared with 85% for the 2D-CNN. This proves that the proposed model is more efficient and generalized, positioning it as a superior choice for embedding in resource-limited devices. Our CNN ensures efficient real-time processing by substantially reducing computational complexity and memory usage, achieving state-of-the-art performance with reduced computational demands. This balance of efficiency and effectiveness establishes the proposed CNN as an ideal candidate for applications requiring minimal computational load and real-time processing capabilities, outperforming not only the 2D-CNN but also the other authors' models, thereby presenting an effective solution for embedding within real-time activity recognition systems.

We also compared the proposed model to the DL models as implemented according to the network configurations and hyperparameter settings detailed in [15] and [26] on unseen testing data from their dataset [15]. The results, shown in Table XI, indicate that our classifier significantly outperformed the state-of-the-art approaches.

VI. CONCLUSION

In this article, we reviewed Wi-Fi CSI-based HAR and introduced HAR-LightCNN, a new approach for classifying human-generated raw CSI data. This model, a lightweight CNN, was trained on the existing datasets. We used time series DA to address small-sized dataset and class-imbalance issues. Testing on two subsets showed that DA techniques enhanced classification and generalization, surpassing current methods

such as 1D-CNN, 2D-CNN, LSTM, and BLSTM. Future plans include designing a deep generative neural network DA for these issues and exploring amplitude, phase, and combined amplitude-phase data for activity recognition.

ACKNOWLEDGMENT

The authors express their gratitude to Moshiri et al. [14] and Yousefi et al. [15] for making their activity datasets publicly available, which they used in evaluating their proposed approach.

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